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Original Article

Effects of combined whole body vibration and resistance training on lower quadrants electromyographic activity, muscle strength and power in athletes

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ABSTRACT

Background: Whole body vibration (WBV) with resistance training is one of the increasing ways of gaining ankle and foot complex muscle strength and power for the rehabilitative and prophylactic purpose in athletes.
Objective: The purpose of the study was to compare the effects of combined WBV and resistance training (RVE) with strength training alone (RE) on alteration of gastrocnemius lateralis and vastus medialis obliquus muscle activity and strength, and power performance in athletes.
Methods: The study was performed on 23 university-level male athletes who were randomized into two groups as RVE (n = 12; age 22.2 ± 1.94 years) and RE (n = 11; age 21.60 ± 1.78 years). The training program was scheduled three times per week for six weeks (18 sessions). Gastrocnemius lateralis (GL) and vastus medialis obliquus (VMO) were measured for muscle activity and isometric strength with surface EMG device and hand-held dynamometer respectively. Counter-movement jump (CMJ) was used for measuring power. All the participants were assessed for outcome measures at baseline and then after 6 weeks. Group (RVE vs. RE) by time (pre vs. post) effects were compared through a 2-way interaction utilizing mixed model repeated measure ANOVA.
Results: After training, VMO muscle activity (group effects) increased significantly in the RVE group ($p < 0.05$). However, both the groups showed statistically significant time and group × time interaction effects for muscle activity of VMO, isometric strength (VMO and GL), and CMJ ($p < 0.05$).
Conclusion: WBV might serve as an adjunct modality for enhancement of the neuromuscular activity of the VMO muscle. However, RVE had no additive effect when compared to RE alone on muscle strength and power in athletes. The long-term impacts of combined WBV and resistance training on other foot and ankles muscle should be investigated in future studies.

1. Introduction

Ankle and foot complex muscle strength and power are an important foundational conditioning requirements in multiple sports types; such as volleyball, football, and basketball. Improvements in ankle and foot strength and power can translate into better performance and recovery after injury and are a focus of strength and conditioning programs. Whole body vibration exercise (WBV) is an exercise approach used with the intent of amplifying strength gains, muscle activity, with the aim to improve athletic performance [1,2]. In WBV the participant stands on a platform that delivers a mechanical vibration stimulus to the entire body, via the feet, in the form of oscillatory motion. WBV generates reflex activity in the muscles by activating the α-motor neurons using mechanical vibrations thereby stimulating the primary nerve endings of muscle spindles and mechanoreceptors in the skin, ligaments, tendons [3–7]. Prior studies have elaborated both the acute and chronic effects of WBV training alone in rehabilitation, bed rest, and athletic training [8, 9]. The initial spurt of gains in muscle strength and power are thought to be due to neural adaptations [10]. A greater increase in muscle strength and power when untrained adults, underwent combined WBV and resistance training as compared to resistance training alone [11–13]. The additive effect generated by WBV and resistance training is thought to be due to elicitation of the tonic vibration reflex (TLR) [3]. This reflex promotes greater muscular co-contractions that may present a greater

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stimulus for adaptation of muscle strength and power. WBV frequency is also considered important and the overload principle using a gradation of vibration frequency is required to elicit a response in participants [12]. In another study comparing the effectiveness of low- and high-frequency WBV, the authors reported that low-frequency WBV was more effective in improving strength and counter movement jump height (CMJ) [14]. CMJ is a reliable and valid measure used in athletes to measure lower-limb power gains [15]. The muscles in the human body are categorized as slow-twitch and fast-twitch fibers [16] and the pattern of recruitment of the muscle while jumping follows a proximal-to distal activation sequence in football, basketball, and volleyball. This sequential recruitment specifically targets the lower leg extensors with an emphasis on the vastus medialis oblique (VMO) (fast-twitch fibers) and gastrocnemius lateralis (GL) (slow-twitch fibers) among other muscles. For this reason, we consider it relevant to examine VMO and GL in analyzing the effects that WBV in CMJ performance [17]. Both VMO and GL are clinically useful in the prevention of various sporting injuries because of their specific contribution in jumping and sprinting activity in the selected sports [18]. A weakness of these muscles had been previously found to be correlated with an increased risk of pain, dysfunction, and pathological changes in the patellar cartilage [19]. An increase in the patellar lateral tracking and patella-femoral joint pressure had also been observed thereby necessitating the formulation of strategies for enhancing the efficiency of these muscles [19]. WBV lowers the excitation threshold of the central and peripheral nervous system which augments the preferential recruitment of the fast-twitch fibers [20]. Due to exercise training, athletes develop neuromuscular adaptations such as reversal of motor fiber recruitment (fast-twitch) and muscle strength [21]. However, there is a dearth of evidence from studies combining resistance training and WBV in athletes and hence the effects on the VMO and GL muscle are less clear. The result of studies done on other non-athletic populations cannot be generalized to athletes as there are existing neuromuscular adaptations in athletes due to their sports-specific training and hence it becomes imperative for research to be done on the athletic population. Volleyball, basketball and football involve running, jumping and cutting movements thereby presenting with a higher proportion of ankle injuries [22,23]. Lower extremity muscle weakness had been suggested as a contributory factor for ankle injuries [24] and the role of strengthening the ankle musculature had been put forth to reduce this injury risk. Previous unstable surface training used for ankle musculature had demonstrated effectiveness in terms of reducing the lower extremity injuries [25].

The purpose of this study was to compare the combined effects of WBV and resistance training (RVE) on muscle activity (VMO and GL), muscle strength (VMO and GL), and CMJ with resistance training alone in athletes. To date, there were several studies that used WBV but the current evidence is lacking in terms of studies that had used a progressive WBV protocol for trained multi-sport players and most of the previous literature is non-progressive or unequivocal in terms of progression [4,26–33]. This is the first study that has analyzed conditioned athletes from multiple sports for several sports-specific and reliable outcome measures following a progressive WBV training and progressive strength training for 6 weeks.

2. Methods and materials

2.1. Study design

A randomized controlled trial with the two-arm parallel design was conducted at the physiotherapy clinic of Jamia Millia Islamia University, with an evaluation conducted at baseline and six weeks following interventions (total 18 treatment sessions). Institutional Ethical Committee of the university approved the study vide no. IEC reference: 26/11/269/JMI/IEC/2019. The study was performed as per the Declaration of Helsinki criteria [34]. Informed written consent was obtained from participants and they were explained about the purpose of this study and about procedures to which they would be subjected.

2.2. Participants

The subjects primarily participating in football, basketball and volleyball were recruited from the sports complex of the university. Although these athletes are physiologically different but all these sports carry a common component of resistance training in their training regimen [35–38]. The inclusion criteria were male university-level athletes in the age group of 18–30 years, working knowledge of English language, normal BMI range, at least 1 year of playing experience, and involved in sport-specific training on at least 2 days per week. The subjects were excluded if they exhibited the presence of any cardiovascular, respiratory, abdominal, urinary, neurological, or musculoskeletal diseases, prior intervention/training on WBV in the last 3 months, any lower extremity surgery in the past 1 year, ocular diseases affecting the retina, history of epilepsy [39]. The sample size for the study was calculated with the help of the G*Power software version 3.1 [40]. The calculations were based on finding a difference in primary outcome measure (muscle activity), at an alpha level of 0.05 and power of 0.80. This generated a total sample size of 23 subjects. A computer-generated randomization number was used to allocate athletes to the two intervention protocols combined WBV training plus resistance training (RVE) and resistance training (RE) alone. These 23 university-level male players (football, basketball, and volleyball) were randomly divided into RVE group (n = 12), (age = 22.2 ± 1.94 years; BMI = 22.63 ± 1.60 kg/m^2) and RE (n = 11), (age = 21.60 ± 1.78 years; BMI = 23.02 ± 1.05 kg/m^2) and measured for descriptive details (Table 1) (Fig. 1). There was no significant difference between the mean values of age, body mass, height, BMI, and leg length between the two groups at baseline ($p > 0.05$).

2.3. Interventions

Group RE received progressive strength training alone and group RVE received progressive WBV in combination with progressive strength training. The athletes in both groups had to perform training 3 times a week for 6 weeks (total 18 sessions). The study protocols for both groups are described in Table 2.

2.3.1. RE group

Group RE with the progressive strength training protocol consisted of two exercises, viz. squats and heel raise. The subjects were made to perform three sets with 8 repetitions of squats and 3 sets with 12 repetitions of heel raise with a rest interval of 1 min between exercises and 60-s of rest between two consecutive sets. The training load was set at 80% of 1-RM.

2.3.2. RVE group

Group RVE participants underwent the strength training protocol (mentioned above for Group RE) on WBV equipment with the frequency of the equipment being gradually raised from 20 Hz in the 1st week to 40 Hz in the 6th week, with an increase of 5 Hz each week for the first 4

Table 1
Comparison of demographic and baseline characteristics between groups.

Variables	RVE group (n = 12)	RE group (n = 11)	p value
Age (years)	22.2 ± 1.94	21.6 ± 1.78	0.45
Height (m)	1.8 ± 0.05	1.79 ± 0.04	0.79
Weight (kg)	72.95 ± 2.9	73.26 ± 2.76	0.53
BMI (kg/m^2)	22.63 ± 1.60	23.02 ± 1.05	0.49
Leg length (m)	0.94 ± 0.01	0.94 ± 0.01	0.88

m: metres; kg: kilograms; BMI: body mass index; cm: centimeters; RVE: resistance vibration exercise; RE: resistance exercise; SD: standard deviation; P: level of significance.

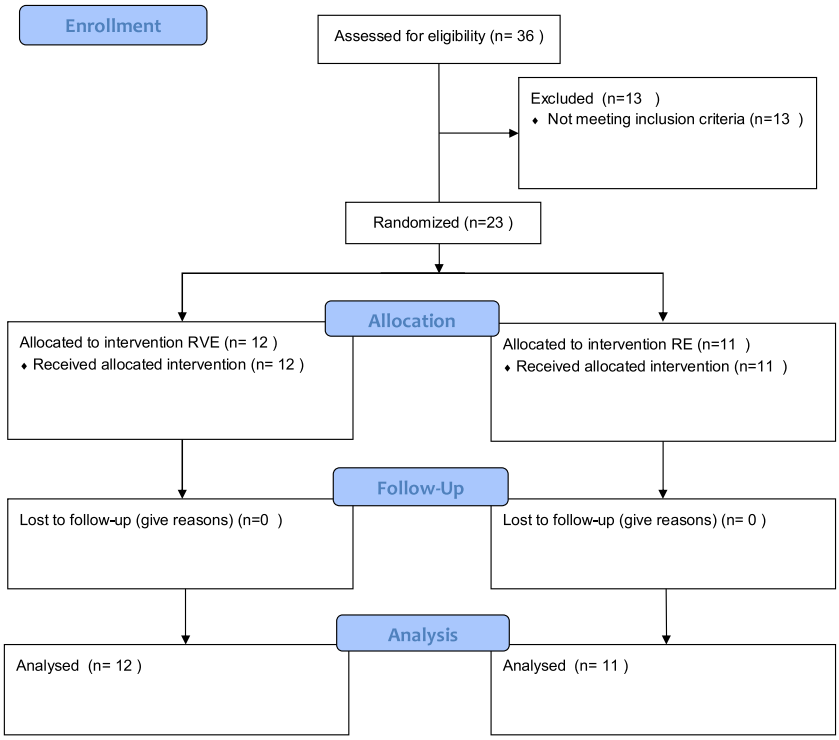

Fig. 1. Consort study flow diagram.

Table 2
Summary of training protocols.

Muscles	Exercises	Dosage	Load and progression
Vastus Medialis Oblique	Standing and squatting till 90° with barbell on shoulders [32].	Frequency: 2 times per week for 2 weeks and 3 times per week for the next 4 weeks Intensity: 3 sets, 1st two sets of 8 repetitions and 3rd set of maximum possible repetitions Duration: 6 weeks	1. RE group 1st session load was set at 80% of 1- RM Progress load by 5% (maximum 10 kg) in subsequent session if able to perform more than 8 repetitions in the 3rd set. Reduce the load by 5% (maximum 10 kg) if less than 8 repetitions are performed in 3rd set.
Gastrocnemius lateralis	Raising heels with a weighted barbell on shoulders [32].		2. RVE group RE protocol + Start with 20 Hz for 1st week progressing 5 Hz every week and using 40 Hz for the last two sessions.

VMO: vastus medialis oblique; RE: resistance exercise; RM: repetition maximum; Hz: hertz; kg: kilograms; cm: centimeters; RVE: resistance vibration exercise; RE: resistance exercise; SD: standard deviation.

weeks and then keeping the same frequency for next 2 weeks (Fig.2a and b).

2.3.3. *Outcome measures*

Measurement and evaluation were done for the following variables at both the baseline and follow-up after 6 weeks:

2.3.3.1. Evaluation of muscle activity. This evaluation of muscle activity was performed with the help of surface electromyography (sEMG) PowerLab 15T (AD Instruments Ltd, Australia). Muscle activity is also termed as Percentage Maximum Voluntary Isometric Contraction (% MVIC). Skin resistance was reduced by rubbing the skin with alcohol gauze. Muscle activity of the Vastus medialis obliques (VMO) and Gastrocnemius lateralis (GL) was recorded using Ag/AgCl disposable electrodes with a clear adhesive hydrogel. The electrodes were then connected to a 2-channel EMG device. Following the correct placement of the electrodes as per the SENIAM guidelines, Vastus medialis oblique (VMO) and gastrocnemius lateralis muscle (GL) activity was inspected by checking the sEMG signals on the computer display [41]. Pre-amplification of the received signal was done at a sampling rate of 1000 Hz and rejection ratio of 115 dB. A bandpass filter was used with a frequency range of 70–300 Hz. Muscle contractions were repeated with a rest period of 5 s controlled by a metronome. This rectification and signal smoothing yielded a peak average EMG value for each trial for a 5 s window. Root Mean Square (RMS) calculation was also performed by asking the participant to perform free physiologic movements. A 5 min rest period was provided following MVIC testing and then the subjects performed extension of the knee during 3 phases (concentric, isometric, and eccentric) with each lasting 3 s. For the gastrocnemius, the subjects were asked to plantarflex the foot. Subjects completed 5 repetitions of these movements, with a rest period of 3 s between consecutive trials. % MVIC calculation was done as per the following equation [42]:

$$\% \text{ MVIC} = (\text{RMS/MVIC}) \times 100$$

MVIC and RMS with EMG testing have been found to be possessing high reliability (ICC = 0.99) for GL muscle [43,44] activation and high

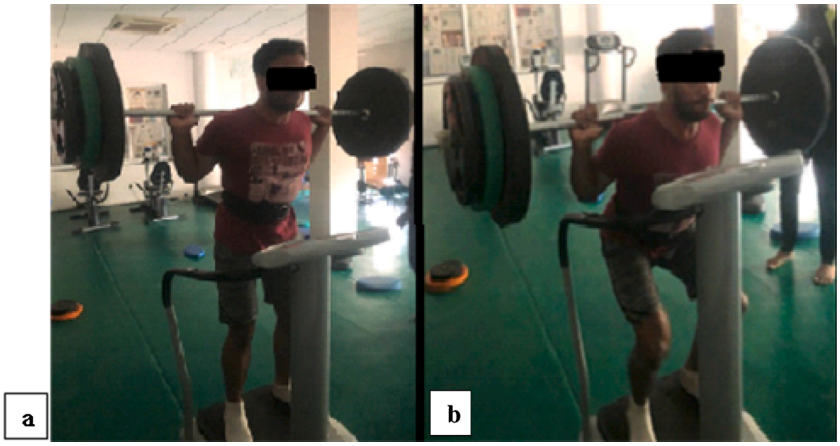

Fig. 2. (a) Subject standing with WBV; (b) Subject performing squats with weights on WBV.

to moderate reliability (ICC = 0.81–1.00) for GL muscle [45,46].

2.3.3.2. Evaluation of counter movement jump (CMJ) height. CMJ is a reliable measure to assess power in the athletic population. The athletes performed it by flexing their knees and hip joints till the point a quarter squat position is reached. Then, the athlete immediately jumped as high as possible by rapidly extending their hip and knee joint along with active flexion of arms to gain vertical momentum [2]. Three trials of the CMJ were taken and the best value out of the three was recorded. Jump height was determined using a measuring tape fastened to a white paper on which each subject's middle finger ink print was recorded. CMJ height has been shown to possess high reliability with an ICC value of 0.98 [47].

2.3.3.3. Evaluation of One Repetition Maximum (1-RM). Before the training protocol, testing was done for 1- RM by performing a warm-up with 5–10 repetitions for each lift at 40%–60% of the perceived maximum. Following a rest period of 1- min including stretching, the subjects performed 3–5 repetitions with 60%–80% of the perceived maximum. The subjects underwent the maximal attempt after a 3−5 min rest interval [42].

2.3.3.4. Evaluation of isometric strength measurement. Subjects performed an isometric-make contraction against a Hand Held Dynamometer (Lafayette Instrument Co, Lafayette, IN, USA) for 5 s followed by determination of peak values for 3 subsequent repetitions. This was followed by a rest period of 10 s between consecutive contractions along with a rest period of 3 min between the agonist group of muscles to allow for alterations in the testing position. Subjects sat at the edge of the table in a high sitting position for assessing isometric knee extensor strength with arms crossed across the chest with the thighs stabilized with straps placed distal to the bilateral hip joint line and the HHD force pad was placed 5 cm proximal to the bimalleolar line. Then, a long sitting position was assumed for the ankle plantar flexors with their arms folded across their chest region, tibia stabilized, and a towel roll kept under the knee with the force pad of the HHD placed at the plantar aspect of metatarsal heads [48]. Hand-Held Dynamometer has been demonstrated to have high intra-rater reliability for measuring knee extensor (ICC = 0.93−0.98) and ankle plantar flexors (ICC = 0.93−0.98) isometric strength performance in an active healthy population [49].

2.4. Statistical analysis

SPSS version 21.0 (SPSS Inc, Chicago, IL) was used for the statistical analysis. The data values were expressed in terms of mean ± standard deviation (SD). Shapiro Wilk test was performed to check for any violation of the assumption of normality. The t-test was used for comparing the baseline intra-group variations in the criterion measures. The mean difference (MD) was calculated by the formula: MD ± t × SD/n. A 2 × 2 mixed ANOVA statistical test was used for the data analysis for within-subject and between-subject effects. Green-house giesser correction was used in case we found a violation of the assumption of sphericity. The effect size or partial eta squared results for 2 × 2 mixed ANOVA with 0.01 (small), 0.09 (medium), and 0.25 (large) was used to interpret the effect size [50]. The level of significance for all tests was set at $p < 0.05$ (Fig.2).

3. Results

The characteristic features of the sample (n = 23) are given in the descriptive statistics in Table 1. Both the groups at baseline were evaluated for outcome measure with unrelated t-test and no significant difference was found ($p > 0.05$) (Table 1). The mean ± SD of the pre and post-intervention and mean difference of the %MVIC VMO, %MVIC GL, isometric strength of VMO, and GL and countermovement jump (CMJ) is calculated and tabulated in Table 3. There were statistically significant main effects for time for %MVIC VMO ($p < 0.01$), %MVIC GL ($p < 0.01$), isometric strength of VMO, GL ($p < 0.01$) and CMJ ($p < 0.01$) (Table 3).

There was a statistically significant group effect for %MVIC VMO ($p < 0.01$) and effect size or partial eta squared depicting large effects. However, there were no statistically significant differences for %MVIC GL ($p > 0.05$), isometric strength of VMO, and GL ($p > 0.05$) and countermovement jump (CMJ) ($p > 0.05$) (Table 3). There were statistically significant interaction effects (time × group) for %MVIC VMO ($p < 0.01$), isometric strength of VMO, and GL ($p < 0.01$) and CMJ ($p < 0.01$) (Table 3).

4. Discussion

Training modalities like WBV are used by athletes to supplement to their training. The main findings of this study showed that the combined resistance training with WBV protocol of 6 weeks significantly increased the VMO muscle activity in the WBV group as compared to resistance training alone in athletes (main effect group).

To the best of our knowledge, this is the first study comparing progressive WBV with progressive strength training for multiple events based (football, basketball, and volleyball) university-level athletes. The combination of conventional resistance training with WBV training has been established to improve muscle strength [10–13]. However, a modification of the progression parameters by sequentially and progressively increasing the frequency and external 1-RM load in the included athletic population is being proposed.

The results observed in terms of improvements in the % MVIC with EMG of VMO were comparable to one of the previous studies that

Table 3
Comparison of outcome measures for both groups before and after the intervention, mean difference and summary of mixed model analysis of variance.

Variables	Time (T)	Group (G)		T		G		G * T	
		RVE (n = 12)	RE (n = 11)	p	E.S	p	E.S	p	E.S
% MVIC VMO	Pre	62.79 ± 7.8	63.64 ± 3.54	<0.01*	0.78	0.01*	0.33	≤0.01*	0.59
	Post	83.68 ± 7.78	68.83 ± 4.89						
	MD	−20.89 ± 7.70*	−5.19 ± 4.20						
% MVIC GL	Pre	65.06 ± 6.07	62.73 ± 7.20	≤0.01*	0.78	≥0.05	0.02	≥0.05	0.03
	Post	78.72 ± 4.92	73.96 ± 6.82						
	MD	−13.66 ± 5.32*	−11.23 ± 7.00*						
VMO HHD (N)	Pre	273.3 ± 41.75	265.71 ± 35.29	≤0.01*	0.98	≥0.05	0.05	≤0.05*	0.42
	Post	345.57 ± 38.57	321.21 ± 38.40						
	MD	−72.27 ± 39.23*	−55.50 ± 36.56**						
GL HHD (N)	Pre	356.4 ± 68.80	353.81 ± 40.30	≤0.01*	0.95	≥0.05	0.01	≤0.01*	0.25
	Post	439.84 ± 69.03	417.97 ± 40.14						
	MD	−83.44 ± 60.23*	−64.16 ± 40*						
CMJ (m)	Pre	0.47 ± 0.05	0.47 ± 0.05	≤0.01*	0.97	≥0.05	0.02	≤0.01*	0.77
	Post	0.55 ± 0.05	0.51 ± 0.05						
	MD	−0.08 ± 0.05*	−0.04 ± 0.05*						

Significant difference: *p < 0.01; **p < 0.05; %MVIC VMO: Percentage Maximum Voluntary Isometric Contraction Vastus medialis oblique; %MVIC GL: Percentage Maximum Voluntary Isometric Contraction gastrocnemius lateralis; VMO HHD: handheld dynamometer isometric strength of Vastus medialis oblique; GL HHD: handheld dynamometer isometric strength of Gastrocnemius lateralis; CMJ: countermovement jump; E.S: effect size or Partial Eta Squared; T: time; G: group; G * T: group * time; MD: mean difference.

compared the EMG following a 13-weeks non-progressive WBV protocol with or without a resistance training protocol [51]. However, several other studies reported conflicting findings as they showed that the WBV training did not induce an increase in % MVIC of knee extensors following 9-weeks of squat, progressive WBV, or combination training [4]. Duc et al. also reported no additional gains in the EMG activity of VMO following an acute bout of non-progressive WBV as part of preconditioning exercises in cyclists [52].

The non-significant improvements in GL muscle activity is in concordance with the results reported in the previous studies. Rosenberger et al. reported a decrease of EMG of GL following 5-days of non-progressive WBV training [7]. In coherence with our study, no significant difference was observed in the GL % RMS in an acute WBV training at 40 Hz [52]. However, an increase in GL % MVIC was observed during various squat variations on a vibratory platform at 35 Hz as compared to the control group [53].

For the CMJ height, both groups demonstrated improvements but no significant difference was observed between the two groups at the end of the intervention. Supporting our results, Rosenberger et al., showed no difference between groups for CMJ height at a 6-week follow-up [29]. However, two previous studies showed significant improvements in the CMJ height following 12 and 24 weeks of WBV training [11,54]. Compared to Rosenberger et al., the WBV training in the other two studies was conducted for a longer period (6 weeks vs. 12 weeks vs. 24 weeks respectively) [11,29,54]. A previous meta-analysis showed WBV training to be superior in terms of increasing CMJ height. However, the RCTs included lacked in terms of symmetry and methodological consistencies as the participants included in the meta-analysis were of varying characteristics [55]. The above findings suggest that the results of WBV training have a population-specific feature.

No differences were observed in the isometric strength of VMO between the two groups at the end of the intervention. For the WBV intervention, Bogaerts et al. assessed isometric knee extensor strength and compared this to a combination of conventional forms of exercises (resistance, aerobic, balance, and flexibility exercises), with no additional improvements seen between groups [56]. Another study utilizing 11 weeks of non-progressive WBV training showed no added benefits in the knee extensor strength [57]. However, a previous RCT utilizing progressive WBV for 12 weeks demonstrated added effectiveness of WBV for improving isometric knee extensor strength [11]. The conflicting results might be attributed to the longer time duration used in comparison to the 6 weeks protocol in our study.

No additional benefits of progressive WBV were observed in the isometric strength of GL muscle. Similar results have been observed in a previous RCT that compared acute non-progressive WBV training to sham intervention (exercises without vibration stimulus) [58]. In contrast to our findings, few studies demonstrated an increase in isometric plantar flexor strength following 6 weeks of progressive and 8 weeks of non-progressive WBV training [29,59]. The conflicting findings might be attributed to the untrained participants included in these studies. Our study included trained university-level athletes demonstrating a smaller tendency of improvement owing to the previous levels of training as compared to untrained individuals [10]. This may be a plausible reason as to why our study did not affect muscle isometric strength of VMO and GL in athletes despite it being shown that similar protocol in non-conditioned subjects induced significantly improved performance. However, this latter group in the mentioned literature had more scope of showing improvement as the neuromuscular apparatus was relatively deconditioned [13].

Secondly, the repetitive stretch-shortening cycle during athletic events stiffened the muscle-tendon unit in comparison to unconditioned subjects. The alteration in leg muscle activity in the pre-landing phase helps to fragment muscle compliance from eccentric loading and vibration-induced damage during the landing [60,61]. Due to this, the muscle-tendon complex of athletes is non-compliant to length alterations and absorption of the vibrations. The main principle behind WBV is that it should be able to elicit sufficient vibration which should overload the muscle complex. Overloading of the muscle results in strength gains which probably didn't happen in our study for VMO and gastrocnemius lateralis and this could be attributed as a possible cause of the obtained result. Therefore, a tailor-made WBV program that incorporates higher oscillatory loadings of the plate might be necessary to bring about at least an equivocal adaptive response in outcomes measures of athletes [62].

The gradually increasing scientific evidence demonstrates proximal muscular changes in patients with ankle dysfunctions. In those with chronic ankle instability, ipsilateral strength deficits have been reported in knee extensors muscle testing [63]. Studies have also shown a reciprocal relationship between gastrocnemius and knee extensors (VMO) with respect to motor neuron pool [64,65]. The current findings add to the available body of knowledge showing the changes in the VMO after RVE training in university-level athletes. This information can be extrapolated for training purposes and athletes undergoing rehabilitation. Clinicians can incorporate the RVE regime in the treatment of athletes with ankle dysfunction or instability for increased activation of the inhibited VMO for lower quadrant stabilization during

weight-bearing activities. Clinicians should view these issues from a wider perspective and design rehabilitation programs for the entire lower extremity of the athletes afflicted with ankle instability or dysfunction. Since our research comprised conditioned athletes, our chosen protocol for the WBV training program may not have been potent enough to cause potential adaptations. The modifiable components of WBV viz. vibration frequency, amplitude, type of exercises, and biaxial vibration platform could have had a significant bearing on the result of the study. Additionally, the duration of the training period is also important. It is scientifically rationalized that in conditioned athletes a longer time frame of training is required for significant improvement to be measured in other outcome measures as well.

A strength of the current study is that this is amongst few studies which have examined a combination of WBV and resistance training in athletes and secondly this study used the progressive increase in vibration frequency which has not been used as frequently in previous studies. The limitations of our study are that our protocol duration was of 6 weeks for trained athletes that could be considered slightly less considering their already existing neuromuscular adaptations and secondly we had taken only male athletes. Thirdly, the two channel s EMG we used in this study limited our ability to monitor only few muscle sites and provides limited understanding about neuromuscular system. Additionally, cross-talk is also known limitation for s EMG measurements where the energy from one muscle group travels over the recording field of another muscle group. This reduces the specificity of the surface EMG in spite of following the standardized electrode placement guidelines. Another shortcoming of s EMG is that electrodes are not totally unobtrusive. The electrodes and leads can potentially encumber a movement pattern and the recording might not reflect the customary pattern of athletic activity. Furthermore, the participants were not blinded to the study groups because of the exercise interventions used. Lastly, we did not analyze the temporal phase of the CMJ that might have provided further insights into lower limb muscle performance. Long-term studies with a duration of around 12 weeks are needed to investigate the effects of combined WBV and resistance training and also examine the functional benefits of an increase in VMO muscle activity which we reported in this study.

5. Conclusion

In conclusion, this study is amongst the first to examine the effects of combined WBV and resistance training. The study found a large increase in VMO muscle activity after 6 weeks of combined WBV and resistance training protocol. However, the study could not find the additional benefit from a combination of WBV and resistance training on other outcome measures. This information may be important for clinicians when designing rehabilitation programs for individuals with functional ankle instability and for those who are at risk for developing an ankle injury.

Conflict of interest

No conflict of interest was associated with this study.

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